Energy Audit Report

Homeowner(s): Jackie Lola

Address: 106 Thunder Road Perry, Maine, 04667

Owner Contact: NA

Auditors: Helen Lloyd, Tyler Hebert, Raquel Longfellow

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Report Date: July 11, 2025

We conducted an energy assessment of your home on 7/10/2025. This report will tell you what we did, what we found, and what we suggest for your home. These suggestions include information on incentives and financing to make improvements more affordable.

Add here the table of content: in Word go to references and add table of content, in Docs go to the insert and on the bottom add the table of context

Summary of your Audit

Visual Inspection and Measurements

We started with a tour and visual inspection of the inside and outside of the home. We identified any visible damage to the building, moisture control strategies, major appliances, and insulation. We measured square footage and volume of the home, as well as the area of all exterior windows and doors. We used a kill-a-watt meter to measure the electricity use of some appliances. During your audit, we used a carbon monoxide meter to measure the ambient carbon monoxide levels throughout the home.

Attic

We entered the attic to check for insulation, air sealing, ventilation, and potential hazards such as mold. Additionally, we visually inspected the attic ventilation and any duct and pipework passing through the attic.

Basement

We visually inspected any appliances in the basement and noted insulation levels, moisture, rodents, and any other concerns.

Blower Door / Air Leakage Test

A blower door test measures the air tightness of a building. We used a specialized fan in an exterior door to generate negative pressure inside your house. The resulting pressure difference between the inside and outside of the house allows us to measure air leakage. To locate the leaks, we used an infrared camera to check for unusually hot and cold spots. We also checked the pressure differences of the rooms to help determine major air leak locations.

Combustion Appliance Safety

We assessed combustion appliances that burn fossil fuels such as propane, heating oil, or kerosene. These include furnaces, boilers, water heaters, and gas ovens. We visually inspected the combustion appliance(s) in your home, but we were unable to perform combustion safety tests. We also performed gas leak detection tests on your propane appliance(s).

Summary of Recommendations

We recommend the following upgrades for your home. Detailed information about these recommendations and financial resources can be found in other sections of this report. These are organized from the least expensive/most cost-effective and easiest interventions to ones that are more challenging to implement.

Recommendations	Description
Solar	Rooftop solar can supply most or all of your home electrical demands. Contact a solar company for pricing and details specific to your home.
Misc. Air Sealing and Insulation	Air seal and insulate [** insert auditor description - bay window, bulkhead door, chase, kneewall, anything else that is unique to the home and not otherwise covered in another recommendation]
Furnace/Boiler Tune-up	Have the furnace/boiler and flue inspected and adjusted by a licensed professional. This should be available from your fuel delivery company.
High-efficiency Shower Head(s)	Install high-efficiency shower heads to reduce the amount of water and energy to heat the water used when showering.
LEDs	Switch your light bulbs to LED light bulbs. LEDs use 80% less energy than incandescent light bulbs, which can significantly reduce your electricity bill.
Refrigerator	Replace your refrigerator with a new, EnergyStar-certified fridge. Look at the Energy Guide label to compare the energy use of new refrigerators.
Induction Stove/Oven	Induction cooking appliances are more efficient and safer than electric or gas ones. There is no risk of carbon monoxide or other harmful combustion gases, and the surface doesn't heat up without a pot or pan on it.
Gutters	Install gutters and downspouts that divert water at least six feet away from the foundation and to where the ground slopes away from the house.
Bathroom Exhaust Fan(s)	Install bathroom exhaust fans in all bathrooms that do not have them. These should be rated for at least 80 cubic feet per minute (CFM) if there is a shower. We recommend the Panasonic Whisper series or similar fans that don't create excess noise.
Kitchen Exhaust Fan	Install a kitchen exhaust fan to remove cooking fumes and moisture from your home. The fan should be rated for at least 100 cubic feet per minute. A kitchen fan should be ducted to the outdoors. If the ductwork will run through the attic, it should be installed before the attic is air-sealed and insulated.
Vapor Barrier	Install a vapor barrier on the basement floor to stop moisture from entering the basement and house.
Spray Foam Basement/Crawlspace Walls	Install spray foam on the basement/crawlspace walls to prevent moisture infiltration and reduce heat loss.
Attic Air Sealing and Insulation	Air seal the attic and insulate it to at least R-60 (18 inches of loose-fill cellulose insulation).
Continuous Exterior Wall Insulation	Add a continuous layer of insulation and potentially replace the air and moisture barrier once it becomes time to replace the siding.
Electrical Panel Upgrade	Consult with an electrician about replacing your existing electrical panel with a 200 amp panel as you add electrical appliances to your home.
Air Source Heat Pump	Install air source heat pumps for heating and cooling and

Info	Values
Date Built:	1992
Foundation Type:	Crawlspace
Attic:	Batts of Fiberglass , 9 inches
Number of floors:	2
Square footage of conditioned space:	1,535 ft^2
Volume of conditioned space:	11,863 ft^3
Ambient Carbon Monoxide reading:	0

Info	Values
Roof age:	15
Orientation:	Northeast/Southwest
Roof type:	The house has a Asphalt Shingles roof.The roof is in NA condition. The shingles show clear signs of age and weathering. They have not begun to fly away during wind events.
Moisture control:	The house has the following moisture control strategies: NA. These were in NA condition. NA
Siding:	The house has vinyl siding. The siding is in fair condition. The siding is in good condition in most areas. There are some spots that have had cracks or breaks in the material.
Exterior doors:	There are 2 hollow fiberglass exterior doors. These are in excellent condition, they have a total area of 37 ft^2.
Exterior windows:	There are 13 double paned wood windows. These are in fair condition, they have a total area of 148 ft^2.

What We Found

Basics

Exterior

Interior/Living space

Blower Door / Air Leakage Test

{r} #| label: blower door data set up for 3.4 #| echo: false

#old set up wiht manual entry to googlespreadsheet blower_door_leakage <- energy_test_googlesheets|> select(“Equivalent leakage area”)|> mutate(`Equivalent leakage area` = paste0(`Equivalent leakage area`, ” in^2”))

Info	Values
Walls:	The house has platform framing, which means that wall cavities do not extend the full height of the building, but are instead separated by floors. The exterior walls are insulated with Batts of Fiberglass insulation. This insulation is 4 inches thick and in fair condition. This gives the exterior walls an R-value of around CALCU-LATE. A material's R-value tells us how good it is at stopping heat flow, the bigger the number, the better. For comparison, current

CFM50:	1250	CFM50 describes how many cubic feet per minute of air are leaving the hou
ACH50:	6.3	ACH50 tells us how many air changes per hour are taking place in the hous
Equivalent leakage area:	125 in^2 under natural conditions.	This is the area (in square inches) equivalent to all of the air leaks in the ho
ACHnatural:	0.5	Accounting for the volume of the home, this means that the house exchange

#blower__door__leakage <-blower__door__leakage[,-c(1)]

blower__door__leakage <-blower__door__leakage|> pivot__longer(c("Equivalent leakage area"), names__to = "Info", values__to = "Values")

blower__door <- energy__test__googlesheets|> select(c("CFM50", "ACH50","ACHn",))|> #there is another column "Comments about air leakage" pivot__longer(c("CFM50", "ACH50","ACHn"), names__to = "Info", values__to = "Values")

all__blower__door <- blower__door|> rbind(blower__door__leakage)|> mutate(Info = recode(Info, "CFM50" = "CFM50:", "ACH50" = "ACH50:","Equivalent leakage area" = "Equivalent leakage area:","ACHn" = "ACHnatural:"))|> arrange(factor(Info, levels = c('CFM50:', 'ACH50:', 'Equivalent leakage area:', 'ACHnatural:')))

all__blower__door['Description'] <- c("CFM50 describes how many cubic feet per minute of air are leaving the house at 50 pascals of pressure difference (while the blower door is running). For every cubic foot of air that leaves the house, a cubic foot of air enters the house as well. The higher the number, the leakier the house.", "ACH50 tells us how many air changes per hour are taking place in the house at 50 pascals of pressure difference. This value is normalized for the volume of the house and thus allows for comparison between different houses. The higher the number, the leakier the house.","This is the area (in square inches) equivalent to all of the air leaks in the house combined.","Accounting for the volume of the home, this means that the house exchanges –% of its air every hour. Over one day, the house goes through – complete air changes.")

{r} #| label: ACH/CFM/ELA/N-factor calculations #| echo: false

table__3__4 <-clean__audit__data|> select(cfm50)

table__3__4<- as.data.frame(as.numeric(table__3__4))

#ACH50 calculation

table__3__4$ACH50_cal <- as.numeric(clean_audit_data$cfm50) * 60 / as.numeric(clean__audit__data$volume__of__c

#Equivalent leakage area calculation

table__3__4$equi_leak_area_cal <- (as.numeric(clean_audit_data$cfm50) / 10)

#N-factor calculation

n__factor__calculated <- clean__audit__data |> select(exposure__of__the__house, number__of__floors)|> mutate(n__factor__cal = case__when(exposure__of__the__house == "Low" &